Unless explicitly requested by a problem, do not include sketches as part of your proof. You are free to use the result from any problem on this (or previous) assignment as a part of your solution to a different problem even if you have not solved the former problem.

**Problem 1** (2pts)**.** For infinitely many integers $n$, construct a graph $G$ with the following properties:

- $G$ is connected and has $n$ vertices, and
- $G$ is *not* a tree, and
- Every $v \in V(G)$ which is not a leaf of $G$ is a cut-vertex of $G$.

**Solution.** We construct such a graph for any even integer $n \geqslant 6$. Consider the integer $m = n/2$ and the cycle $C_m$ on vertices $\{v_1, \ldots, v_m\}$. We define $G$ by adjoining $m$-vertices $u_1, \ldots, u_m$ to $C_m$, and edges $u_i v_i$. Thus $G$ has $2m = n$ vertices and is connected by construction. Also, $G$ is not a tree, because it contains the cycle $C_m$ (here we use that $n \geqslant 6$, or equivalently, $m \geqslant 3$). Note that each $u_i$ is a leaf of $G$, because the only neighbor of $u_i$ is $v_i$. On the other hand, $\deg_G v_i = 3$ for all $i$; thus, the non-leaves of $G$ are $v_1, \ldots, v_m$. Finally, each $v_i$ is a cut-vertex of $G$, because $u_i$ is an isolated vertex in $G - v_i$. □

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**Problem 2** (1 + 1 pts)**.** Let $G$ be a graph and suppose that $H$ is any spanning subgraph of $G$.

(a) Prove that $\lambda(G) \geqslant \lambda(H)$.

(b) Prove that $\kappa(G) \geqslant \kappa(H)$.

**Solution.**

(a) In case $G \cong K_1$, then also $H \cong K_1$, so $\lambda(G) = \lambda(H) = 0$.

Otherwise, let $S \subseteq E(G)$ be a minimum edge-cut for $G$, thus $\lambda(G) = |S|$. Let us show that $S' = S \cap E(H)$ is and edge cut for $H$. Since $G - S$ is disconnected, we can partition $V(G) = A \sqcup B$ where $A, B$ are non-empty subsets of vertices such that no edge in $G - S$ goes between $A$ and $B$. Since $H$ is a spanning subgraph, we have $V(H) = V(G)$, thus this is also a partition for $V(H)$. Since $E(H - S') \subseteq E(G - S)$, we deduce that no edge in $H - S'$ goes between $A$ and $B$. Hence $H - S'$ is disconnected, which means that $S'$ is an edge cut for $H$, and as a consequence we obtain $\lambda H \leqslant |S'| \leqslant |S| = \lambda(G)$, as desired.

(b) In case $G \cong K_n$, we have $\kappa(G) = n - 1$ since $H$ has $n$-vertices, we know from class that $\kappa(H) \leqslant n - 1 = \kappa(G)$.

Now assume $G$ is not complete, and let $U$ denote a minimum vertex-cut for $G$, so we have $\kappa(G) = |U|$. Since $G - U$ is disconnected, we can partition $V(G) - U = A \sqcup B$ where $A, B$ are non-empty subsets such that no edge in $G - U$ goes between $A$ and $B$. But $H$ is a spanning subgraph, hence the sets $A, B$ also partition $V(H) = V(G)$. Since $E(H - U) \subset E(G - U)$, no edge in $H - U$ goes from $A$ to $B$, hence $H - U$ is disconnected. In other words, $U$ is a vertex-cut for $H$, which implies that $\kappa(H) \leqslant |U| = \kappa(G)$.

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**Problem 3** (1 + 2 pts)**.** Let $G = (V, E)$ be a graph with at least one edge and fix any $e \in E$.

(a) Prove that $\lambda(G) \geqslant \lambda(G - e) \geqslant \lambda(G) - 1$.

(b) Prove that $\kappa(G) \geqslant \kappa(G - e) \geqslant \kappa(G) - 1$.

**Solution.**

(a) Since $G - e$ is a spanning subgraph of $G$, we know that $\lambda(G) \geqslant \lambda(G - e)$ by Problem 2. Notice that $G \not\cong K_1$, because $G$ contains at least one edge. Thus, also $G - e$ is not $K_1$ and we can take a minimum edge cut $S$ for $G - e$, so $\lambda(G - e) = |S|$. But then $S' = S \cup \{e\}$ is an edge cut for $G$, because $G - S' = G - e - S$ is disconnected. Thus $\lambda(G) \leqslant |S'| = |S| + 1 = \lambda(G - e) + 1$, which implies that $\lambda(G - e) \geqslant \lambda(G) - 1$.

(b) We have $\kappa(G) \geqslant \kappa(G - e)$ by Problem 2, because $G - e$ is a spanning subgraph of $G$.

Since $G - e$ is not a complete graph, there is a minimum vertex-cut $U$ for $G - e$, so $\kappa(G - e) = |U|$. Next, we build a vertex cut for $G$ using $U$. We consider two cases:

**Case 1:** $G - U$ is disconnected. Hence, $U$ is a vertex cut of $G$ by definition, and as a consequence we have $\kappa(G - e) = |U| \geqslant \kappa(G) \geqslant \kappa(G) - 1$, as desired.

**Case 2:** $G - U$ connected. Since $U$ is a vertex cut for $G - e$, we know that $(G - e) - U$ is disconnected, which means that we can partition $V((G - e) - U) = A \sqcup B$ where $A$, $B$ are non-empty subsets and no edge in $(G - e) - U$ goes between $A$ and $B$. On the other hand, since $V(G - U) = V((G - e) - U)$, we see that $A$ and $B$ also partition $V(G - U)$, but since $G - U$ is connected, some edge in $G - U$ goes from $A$ to $B$. Moreover, that edge must be $e$, so we have $e = ab$ for some $a \in A$, $b \in B$. Without loss of generality, we can assume that $|A| \geqslant |B|$. We consider two subcases.

**Case 2.1:** $|A| = 1$. Thus algo $|B| = 1$, which implies that $|U| = n - 2$ (here, $n$ is the number of vertices of $G$). But then $\kappa(G - e) = n - 2$, and we know that $\kappa(G) \leqslant n - 1$ always, so certainly $\kappa(G - e) \geqslant \kappa(G) - 1$, as desired.

**Case 2.2:** $|A| \geqslant 2$. In this case, $U' = U \cup \{a\}$ is a vertex-cut for $G$. Indeed $G - U'$ is disconnected, since $V(G - U') = (A \backslash \{a\}) \sqcup B$ where both $(A \backslash \{a\})$, $B$ are non-empty and no edge in $G - U'$ goes between $(A \backslash \{a\})$ and $B$. Hence $\kappa(G) \leqslant |U'| = |U| + 1 = \kappa(G - e) + 1$, or in other words $\kappa(G - e) \geqslant \kappa(G) - 1$, as desired.

□

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**Problem 4** (1 pts)**.** For each non-negative integer $k$, find an example of a graph $G$ with $\kappa(G) = \lambda(G) = 1$, yet there is some vertex $v \in V(G)$ such that $\kappa(G - v) = \lambda(G - v) = k$.

That is to say, the natural analogue of Problem 3 fails when deleting vertices instead of edges.

**Solution.** Let $H$ denote a copy of the complete graph $K_{k+1}$ on vertices $\{v_1, \ldots, v_{k+1}\}$. We know from class that $\kappa(H) = \lambda(H) = (k+1) - 1 = k$. Define $G$ by adding a new vertex $v$, and the edge $vv_1$. Notice that $G$ is connected and $vv_1$ is a bridge, thus $\lambda(G) = 1$. Also, $v_1$ is a cut vertex, because $v$ is an isolated vertex in $G - v_1$, hence $\kappa(G) = 1$. Finally, since $G - v = H$, we have $\kappa(G - v) = \lambda(G - v) = k$, as desired. □

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**Problem 5** (2 pts)**.** Let $G$ be a graph. The $k$-*th power of* $G$ is the graph $G^k$ which has the same vertex-set as $G$ and $uv \in E(G^k)$ iff $d_G(u, v) \leqslant k$.

Prove that if $G$ is a connected graph on at least $k + 1$ vertices, then $G^k$ is $k$-connected.

**Solution.** Denote $V = V(G) = V(G^k)$, and let $n = |V|$, so $n \geqslant k + 1$ by assumption.

If $G^k$ is complete, then $\kappa(G^k) = n - 1 \geqslant k$ by assumption, so $G^k$ is $k$-connected.

Otherwise, let $U$ be a vertex-cut for $G$. We need to show that $|U| \geqslant k$, so assume for a contradiction that $|U| < k$. Since $G^k - U$ is disconnected, there are non-empty subsets $A$, $B$ of $V - U$ such that $V - U = A \sqcup B$ and no edge in $G^k - U$ goes between $A$ and $B$. On the other hand, $G$ is connected,

so $G$ contains $a$–$b$ path for any $a \in A$ and any $b \in B$. Consider a pair of elements $a \in A$, $b \in B$ such that $d_G(a, b)$ is minimum. Take $(a = v_0, v_1, \ldots, v_\ell = b)$ an $a$–$b$ geodesic in $G$. Since $d_G(a, b) = \ell$ and $ab \notin E(G^k)$, we know that $\ell = d_G(a, b) \geqslant k + 1$, thus $\ell - 1 \geqslant k > |U|$. This implies that at least one of the $\ell - 1$ internal vertices $v_1, \ldots, v_{\ell-1}$ must be *not* in $U$. Hence, for some $i \in \{1, \ldots, \ell - 1\}$, we have $v_i \notin U$, so either $v_i \in A$ or $v_i \in B$. In the former, we have $v_i \in A$ with $d_g(v_i, b) = \ell - i < \ell$; in the later, we have $v_i \in B$ with $d_G(a, v_i) = i < \ell$. In any case, this $v_i$ contradicts the minimality of $d_G(a, b)$. Thus it must be $|U| \geqslant k$ for any vertex-cut of $G^k$, hence $G^k$ is $k$-connected. □